

GFOL: Grid-following VSC modelling in RAMSES

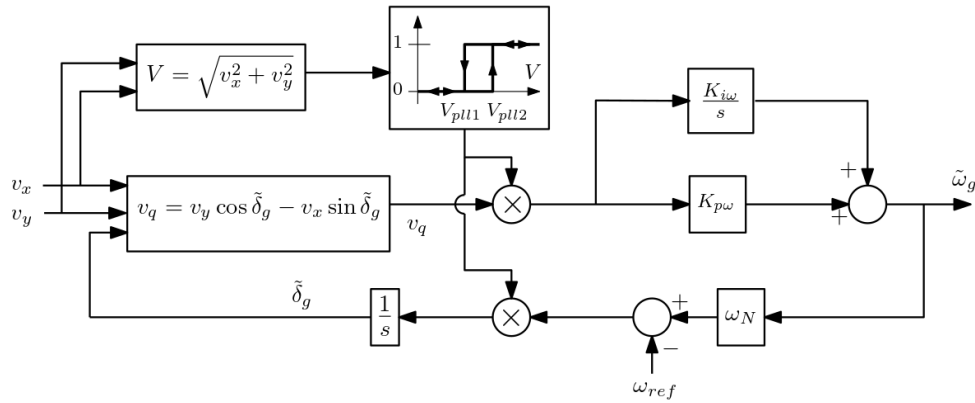
Overview. An MMC-type converter connected to the grid through its transformer (resistance R , inductance L , ideal ratio r ; no LC filter). The DC side is not modelled; the focus is on the AC grid dynamics. The model is rather detailed (measurement low-pass filters, PLL with blocking hysteresis, outer power loops, inner current control loops, dynamic voltage support). Per-unit values refer to the nominal apparent power S_{nom} of the converter; the conversion factor of currents is $k = r S_{base}/S_{nom}$.

Reference frame. The VSC control frame (d, q) tracks the PCC voltage phasor through the PLL angle $\tilde{\delta}_g$:

$$v_d = v_x \cos \tilde{\delta}_g + v_y \sin \tilde{\delta}_g, \quad v_q = -v_x \sin \tilde{\delta}_g + v_y \cos \tilde{\delta}_g$$

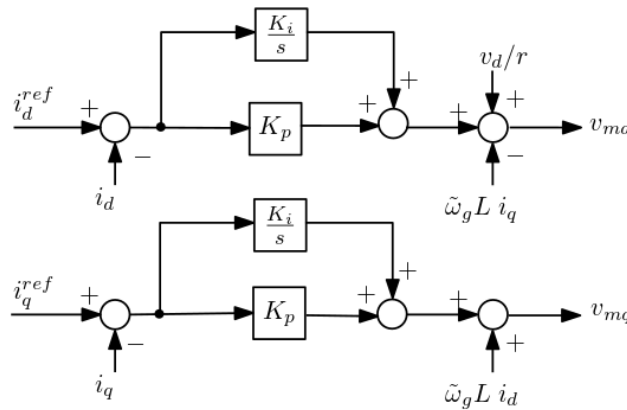
and similarly for the converter-side currents ($i_{xt} = k i_x$, $i_{yt} = k i_y$). In steady state $v_q = 0$ and $v_d = V$.

Phase Locked Loop. The q -axis voltage drives a PI controller whose output is the grid frequency estimate $\tilde{\omega}_g$; integrating $\omega_N(\tilde{\omega}_g - \omega_{ref})$ gives the PLL angle. The gains follow from the PLL response time τ : $K_{p\omega} = 10/(\omega_N \tau)$, $K_{i\omega} = 25/(\omega_N \tau^2)$, $\omega_N = 2\pi f_N$. The PLL is blocked once the PCC voltage V falls below V_{pllb} and reactivated once V recovers above V_{pllu} (hysteresis).



Inner current control. PI controllers with cross-coupling compensation give the modulated voltage:

$$v_{md} = \frac{v_d}{r} - \tilde{\omega}_g L i_q + \left(K_p + \frac{K_i}{s}\right)(i_d^{ref} - i_d), \quad v_{mq} = \tilde{\omega}_g L i_d + \left(K_p + \frac{K_i}{s}\right)(i_q^{ref} - i_q)$$

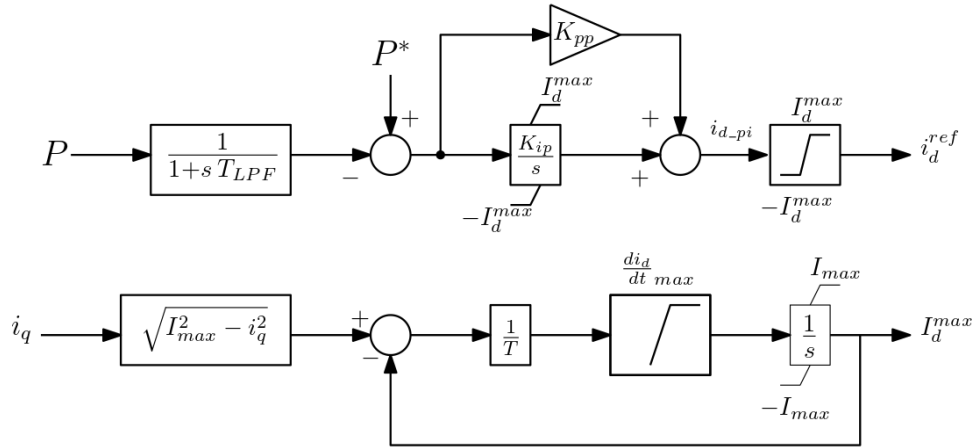


combined with the transformer relationship $v_{md} = v_d/r + R i_d - \tilde{\omega}_g L i_q$, $v_{mq} = v_q/r + R i_q + \tilde{\omega}_g L i_d$.

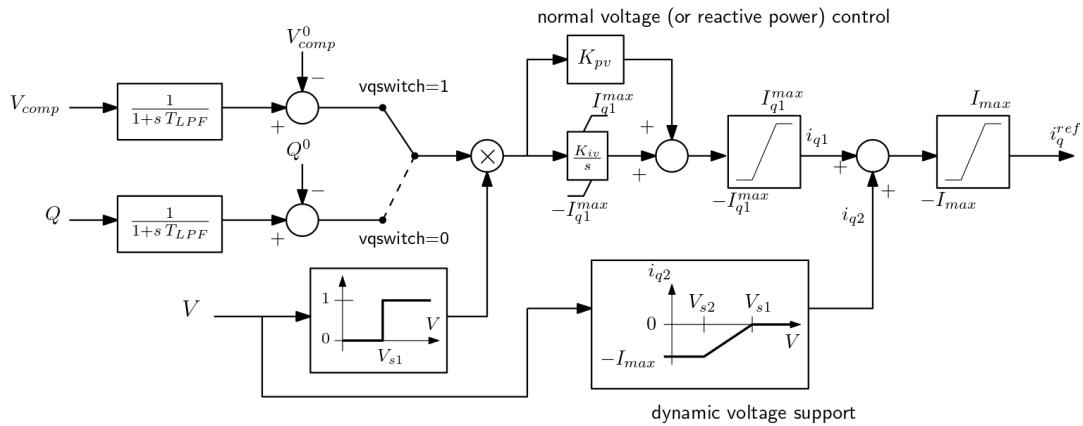
Active power control. The measured power is filtered (T_{lpf}) and compared with the setpoint P^0 ; a PI controller (K_{pp} , K_{ip}) with non-windup limits $\pm I_d^{max}$ produces i_d^{ref} . Priority is given to the reactive current:

$$I_d^{max,stat} = \sqrt{\max(I_{max}^2 - i_q^2, 0)}$$

and I_d^{max} tracks this value through a first-order lag (T_{rlim} , typically 0.002 s) with rate limits dPdt_min/dPdt_max, limiting in particular the rate of recovery of the active current after it has been decreased owing to an increase of i_q .



Voltage / reactive power control. Depending on vqswitch, either the compensated voltage $V_c = |\bar{v}/r + (R_c + jX_c)\bar{i}|$ (vqswitch=1) or the reactive power Q (vqswitch=0) is controlled. The controlled quantity is filtered (T_{lpf}) and its deviation from setpoint drives a PI controller (K_{pv} , K_{iv}) with non-windup limits $\pm i_{q1}^{max}$, giving i_{q1} ; the error is zeroed (controller frozen) while $V < V_{s2}$. Dynamic voltage support adds a component i_{q2} , piecewise linear in V : zero above V_{s1} , decreasing linearly to $-I_{max}$ at V_{s2} , constant below. The total $i_{q1} + i_{q2}$, limited to $\pm I_{max}$, is the reference i_q^{ref} .



With $R_c = R$, $X_c = \omega L$ and $r = 1$, the magnitude of the modulated voltage is controlled.

Data. The record has the following structure (default values correspond to a 1200 MVA converter):

```
INJEC GFOL NAME BUS FP FQ P Q R L r Snom Rc Xc Kp Ki Tlpf Kpp Kip Trlim dPdt_min
dPdt_max Kpv Kiv tau Vpll b Vpllu Imax Vs1 Vs2 iqmax vqswitch ;
```

Parameter	Description	Unit	Default
R	phase reactor resistance	pu	0.005
L	phase reactor inductance	pu	0.15
r	ratio of ideal transformer		1.02
Snom	nominal apparent power	MVA	1200
Rc	resistance used in compensated voltage	pu	0.005
Xc	reactance used in compensated voltage	pu	0.15
Kp	current control: proportional gain		0.573
Ki	current control: integral gain		6.0
Tlpf	measurement time constant (low-pass filter)	s	0.0033
Kpp	active power control: proportional gain		0.0333
Kip	active power control: integral gain		10.0
Trlim	time constant of I_d^{max} rate limiter	s	0.002
dPdt_min	min rate of change of the d -current limit	pu/s	-999
dPdt_max	max rate of change of the d -current limit	pu/s	10.0
Kpv	reactive power control: proportional gain		0.1667
Kiv	reactive power control: integral gain		50.0
tau	response time of PLL	s	0.10
Vpll b	voltage below which the PLL is blocked	pu	0.4
Vpllu	voltage above which the PLL is unblocked	pu	0.5
Imax	maximum current (typically 1–1.2)	pu	1.0
Vs1	voltage below which reactive support starts	pu	-1000
Vs2	voltage of maximum reactive support ($V_{s2} < V_{s1}$)	pu	-2000
iqmax	limit ($\pm$) on quadrature current i_{q1}	pu	1.001
vqswitch	1 = voltage control, 0 = reactive power control		1

Setting Vs1 and Vs2 to large negative values (as in the defaults above) disables the dynamic voltage support. Typical values of Kiv are 50 pu/s for voltage control and 10 pu/s for reactive power control. The setpoints P^0 , Q^0 and V_c^0 are computed at initialization from the power-flow solution and can be modified during the simulation with CHGPRM disturbances.

Example, spanning two lines:

```
INJEC GFOL HVDC1 A 1. 1. 0. 0. 0.005 0.15 1.02 1200. 0.005 0.15 0.5730 6. 0.0033
0.0333 10.
0.002 -999. 10. 0.1667 50. 0.10 0.4 0.5 1.000 -1000 -2000 1.001 1 ;
```

Observables: P_MW, Q_Mvar, Vc, Vm, vmd, vmq, I, w_pll, theta_pll_deg, mult_pll, Idmax, Md, idref, id, id_pi, iq_1, iq_2, iqref, iq.